

SAMPLE PAPER – SET 1

I Yr. B.TECH. (BM, CE, CS, EE & EC Branches) PH10009: APPLIED PHYSICS

Time: 3 Hours

Max Marks: 70

TOTAL SECTIONS: 05 NOTE: Missing data may be assumed appropriately.

Given: $h = 6.626 \times 10^{-34}$ Js, $\hbar = 1.054 \times 10^{-34}$ Js, $k_B = 1.38 \times 10^{-23}$ J/K, $m_e = 9.11 \times 10^{-31}$ kg, $c = 3 \times 10^8$ m/s

Q.No	QUESTION	Marks	CO	BL
90	(a) With neat energy level diagrams, explain the three types of interaction of light with matter: absorption, spontaneous emission, and stimulated emission.	2	1	2
	(b) Match the following: (i) Einstein's A_{21} coefficient (α) Condition for lasing action (ii) Metastable state (β) Coefficient of spontaneous emission (iii) Einstein's A_{21} coefficient (γ) Long-lived excited state (iv) Einstein's B_{21} coefficient (δ) Controls stimulated emission	2	1	1
	(c) (i) What is population inversion and why is it essential for laser action? (ii) Why does a two-level laser not work in practice? (iii) Why are photons called Bosons?	3	1	3
	(d) (i) Derive the relation between Einstein's A and B coefficients by considering thermal equilibrium between matter and radiation. Hence show $A_{21}/B_{12} = 8\pi h\nu^3/c^3$ and $B_{12} = B_{21}$. (ii) A He-Ne laser emits at $\lambda = 6328 \text{ \AA}$ with power 5 mW. Estimate the number of photons emitted per second. (iii) For a Ruby laser, the metastable level has lifetime 3×10^{-3} s. Using HUP, find the minimum energy uncertainty of this level.	3 + 2 + 2	1	4
	OR			
	(e) With energy-level diagrams and neat schematics, explain the construction and working of: (i) Ruby laser (3-level, solid state) and (ii) He-Ne laser (4-level, gas). Compare their active medium, pumping method, output mode, and efficiency.	$3\frac{1}{2} + 3\frac{1}{2}$	1	3
90	(a) Classify optical fibres based on (i) refractive index profile and (ii) number of modes. Draw neat diagrams showing the index profiles for each type.	2	2	2
	(b) Draw a neat ray diagram of light guiding in an optical fibre. Derive an expression for the maximum angle of incidence i_{max} at the input face for no-loss light guiding.	2	2	3
	(c) Write a note on the contributions of any two of the following scientists: (i) N.S. Kapany (ii) S.N. Bose (iii) CKN Patel (iv) ECG Sudarshan	3	2	2
	(d) (i) A ray of light is launched at 7.5° at the air-core interface of an optical fibre of $n_{core} = 1.5$. For TIR find the minimum value of $n_{cladding}$. (ii) The acceptance angle of an optical fibre is 30° when kept in air. Find the acceptance angle when the same fibre is immersed in a liquid of refractive index 1.33. (iii) The attenuation in an optical fibre is 3.6 dB/km. What fraction of its initial intensity remains after 1.5 km?	3 + 2 + 2	2	4
	OR			
	(e) (i) Draw the refractive-index profiles of step-index and graded-index optical fibres. Compare their dispersion and bandwidth characteristics. (ii) If n_0 , n_{co} , and n_{cl} are the refractive indices of air, core, and cladding respectively, show that it is possible to have $NA > 1$ when $n_0 > 1$. What does this physically mean?	3 + 4	2	3

Q.No	QUESTION		Marks	CO	BL
90	(a)	Is the Earth an inertial frame of reference? What about the Sun? Explain. State the two fundamental postulates of Special Theory of Relativity.	2	3	2
	(b)	Find the kinetic energy (KE) of an electron moving at $v = 0.99c$ using (i) classical theory and (ii) relativistic theory. Compare both values and comment.	2	3	4
	(c)	How fast would an athlete need to be running for a 100 m race track to appear only 60 m long? What is unreasonable about this? Discuss.	3	3	5
	(d)	(i) Derive expressions for the relativistic mass-energy relation ($E = mc^2$) and relativistic momentum. Also reduce the expressions for energy and momentum of a massless particle. (ii) A spaceship moving away from Earth at $0.6c$ fires a rocket forward at $0.7c$ relative to itself. What is the speed of the rocket as seen from Earth? (iii) At what speed does a particle's relativistic mass become three times its rest mass?	3 + 3 + 1	3	4
OR					
	(e)	(i) Write the Galilean and Lorentz transformation equations in x, y, z, t . Using Einstein's postulates, derive the Lorentz transformation equations from the Galilean transformation. (ii) The half-life of a pion at rest is 1.8×10^{-8} s. If a pion moves at $0.995c$, find: (a) its half-life as measured by a ground observer, (b) the mean distance it travels before decaying.	2 + 5	3	4
90	(a)	Estimate the number of photons per second emitted from a light source of wavelength 580 nm and optical power 10 W.	2	4	3
	(b)	An atom in an excited state stores energy temporarily. If the lifetime of this excited state is measured to be 100 ps, what is the minimum uncertainty in the energy of the state?	2	4	4
	(c)	An electron and a photon of wavelength 500 nm have the same momentum. Calculate the energy of both the electron and the photon and compare.	3	4	4
	(d)	(i) What is Compton effect? Derive an expression for the wavelength of the scattered photon in Compton scattering. (ii) Incident X-rays of wavelength 10.0 pm are scattered at a certain angle. The scattered X-ray wavelength is 10.5 pm. Find the momentum (magnitude and direction) of the recoiling electron. (iii) What is de Broglie's hypothesis? Calculate the de Broglie wavelength of an electron accelerated through 100 V.	1 + 3 + 3	4	4
OR					
	(e)	(i) What is the significance of the normalisation condition for a wave function? If $\psi = Ae^{-i(Et/\hbar + \alpha x/2)}$ is normalised over $[0, L]$, find A . (ii) Using Schrödinger's time-independent equation, derive the normalised wave functions and energy eigenvalues for a particle in a 1D infinite potential well of width a . Plot $ \psi_n ^2$ for $n = 1, 2, 3$.	2 + 5	4	5
	(a)	What are unitary operations? Does any classical gate have unitary property? Show that the Hadamard gate H is unitary.	2	5	3
	(b)	What do you understand by quantum "entanglement"? Write all four Bell states for a two-qubit system. How are they generated?	2	5	2

90UNIT 5: QUANTUM COMPUTING

Q.No	QUESTION	Marks	CO	BL
(c)	In the following cases, find $ \psi\rangle$: (i) $ 0\rangle \xrightarrow{H} \psi\rangle$ (ii) $ 1\rangle \xrightarrow{H} \psi\rangle$ (iii) $ 0\rangle \xrightarrow{X} \xrightarrow{H} \psi\rangle$ (iv) $ 1\rangle \xrightarrow{H} \xrightarrow{Z} \psi\rangle$	3	5	3
(d)	Solve the following 2-qubit quantum circuit for all possible inputs $ \psi_1\psi_2\rangle$ and present the outputs $ \phi_1\phi_2\rangle$ in the form of a Truth Table. Identify which outputs are Bell states. $ \psi_1\rangle - [H] - \bullet - [X] - \phi_1\rangle$ $ \psi_2\rangle - [X] - \oplus - [H] - \phi_2\rangle$ (The $\bullet\text{--}\oplus$ symbol denotes CNOT with $ \psi_1\rangle$ as control qubit.)	7	5	5
OR				
(e)	Find the output for all possible 2-qubit inputs $ \psi_1\psi_2\rangle$ in the circuit below. Explain the output qubits in each case and show whether the outputs are entangled states. $ \psi_1\rangle - [H] - \bullet - \phi_1\rangle$ $ \psi_2\rangle - [H] - \oplus - \phi_2\rangle$ Also prove: (i) $H = (X + Z)/\sqrt{2}$ and (ii) $H^2 = I$, using matrix representations.	7	5	5

CO – Course Outcome. BL – Bloom’s Taxonomy: 1-Remember, 2-Understand, 3-Apply, 4-Analyse, 5-Evaluate, 6-Create.

SAMPLE PAPER – SET 2

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Q.No	QUESTION	Marks	CO	BL
90	(a) Explain the characteristics of stimulated emission that make it the basis of laser action. What makes laser light different from ordinary light (discuss monochromaticity, coherence, and directionality)?	2	1	2
	(b) Match the following: (i) Ruby laser (α) 4-level continuous-wave gas laser (ii) He-Ne laser (β) 3-level pulsed solid-state laser (iii) Brewster's window (γ) Eliminates unwanted polarization modes (iv) Optical resonator (δ) Converts active medium into oscillator	2	1	1
	(c) (i) Why is the He-Ne laser preferred over the Ruby laser for continuous-wave applications? Explain using the concept of metastable states and energy levels. (ii) What is the role of liquid nitrogen cooling in the Ruby laser?	3	1	3
	(d) (i) Starting from the condition for thermal equilibrium between radiation and matter (Boltzmann statistics + Planck's formula), derive the complete relation between Einstein's A_{21} , B_{21} , and B_{12} coefficients. Show $B_{12} = B_{21}$ and $A_{21} \propto \nu^3$. (ii) The ratio of stimulated-to-spontaneous emission at $T = 300$ K: calculate this ratio for $\lambda = 500$ nm. What does this imply for laser operation at room temperature? (iii) A Ruby laser emits a pulse of 1 J energy in 10 ns at $\lambda = 694.3$ nm. Find the peak power and number of photons in the pulse.	3 + 2 + 2	1	4
	OR			
	(e) Write a detailed essay on the contributions of any two scientists from the list below to the field of Lasers and Photonics. Include their key discoveries, the scientific principles they established, and at least one quantitative formula or result associated with each. (i) S.N. Bose (ii) CKN Patel (iii) M.L. Bhaumik (iv) ECG Sudarshan	$3\frac{1}{2} + 3\frac{1}{2}$	1	2
90	(a) What is Total Internal Reflection (TIR)? How is it the fundamental principle governing light propagation in an optical fibre? What is the critical angle and how is it defined?	2	2	2
	(b) Define V-number (normalised frequency) of an optical fibre. An optical fibre has core radius $a = 30 \mu\text{m}$, $\mu_1 = 1.50$, $\mu_2 = 1.48$, at operating wavelength $\lambda = 0.85 \mu\text{m}$. Calculate the V-number and state whether it is single-mode or multimode.	2	2	3
	(c) Write a detailed note on the contribution of N.S. Kapany to the field of fibre optics. Why is he called the "Father of Fibre Optics"?	3	2	2
	(d) (i) Derive the expression for Numerical Aperture (NA) of a step-index optical fibre. Starting from Snell's law at the launching face and at the core-cladding interface, show that $\text{NA} = \sqrt{\mu_1^2 - \mu_2^2}$. (ii) A ray is incident at 8.5° at the input face of an optical fibre with $n_{\text{core}} = 1.6$. Find the minimum n_{cladding} for TIR. (iii) The acceptance angle of a fibre in air is 22° . When immersed in liquid of refractive index 1.35, find the new acceptance angle.	3 + 2 + 2	2	4

Q.No	QUESTION	Marks	CO	BL
OR				
	(e) (i) Explain the different types of losses that occur in optical fibres: absorption loss, scattering loss, radiative (bending) loss, and dispersion. Give the expressions where applicable. (ii) Draw and explain the complete block diagram of a fibre-optic communication system. State four important advantages of this system over conventional copper-wire communication.	3 + 4	2	3
90	(a) Distinguish between inertial and non-inertial frames of reference with examples. State clearly: is the surface of the Earth a perfectly inertial frame?	2	3	2
	(b) Show that for $v \ll c$, the relativistic kinetic energy expression $K = (m - m_0)c^2$ reduces to the classical result $K = \frac{1}{2}m_0v^2$.	2	3	3
	(c) A proton moves at $0.998c$. Find its (i) relativistic mass, (ii) relativistic momentum, and (iii) total energy. [Rest mass of proton $= 1.67 \times 10^{-27}$ kg]	3	3	4
	(d) (i) Using Lorentz transformation equations, derive the relativistic velocity transformation for all three components u'_x, u'_y, u'_z . (ii) A rocket moving at $0.80c$ fires a projectile in the forward direction at $0.60c$ relative to the rocket. What is the speed of the projectile as measured by a ground observer? (iii) A proper-length rod of 5 m is inclined at 45° to the direction of motion and moves along x -axis at $v = 0.6c$. Find the apparent angle of inclination and apparent length as seen by a stationary observer.	3 + 2 + 2	3	4
OR				
	(e) (i) Derive the energy-momentum relation $E^2 = (m_0c^2)^2 + (pc)^2$ from the definitions of relativistic energy and momentum. Reduce it for a massless particle. (ii) Explain the Twin Paradox of STR. How is it resolved without contradicting the principle of relativity? (iii) An electron and positron, each with kinetic energy 1.5 MeV, annihilate. Find the energy and wavelength of the gamma photon produced. [$m_e c^2 = 0.511$ MeV]	3 + 2 + 2	3	4
90	(a) State Planck's radiation formula. Show how it reduces to (i) Wien's displacement law at high frequency and (ii) Rayleigh-Jeans law at low frequency.	2	4	3
	(b) A photon and an electron have the same de Broglie wavelength of 0.20 nm. Compare their momenta, speeds, and kinetic energies.	2	4	4
	(c) X-rays of wavelength $0.50 \text{ \AA}$ are scattered through 60° by a free electron. Find: (i) Compton shift, (ii) wavelength of scattered photon, and (iii) kinetic energy of the recoil electron.	3	4	4
	(d) (i) What is the "ultraviolet catastrophe"? How did Planck's quantum hypothesis resolve it? Derive Planck's average energy expression $\bar{E} = h\nu / (e^{h\nu/k_B T} - 1)$ using Maxwell-Boltzmann statistics. (ii) If $\psi = Ae^{-i(Et/\hbar + \alpha x/2)}$ is a normalised wave function over $[0, L]$, find the value of A . (iii) The position of a 0.5 keV electron is measured with accuracy $\Delta x = 0.1$ nm. Find the minimum uncertainty in its velocity and the percentage uncertainty in its momentum.	3 + 2 + 2	4	4
OR				

Q.No	QUESTION	Marks	CO	BL
	(e) (i) Starting from the classical wave equation and using the de Broglie relation and energy conservation, derive Schrödinger's time-dependent wave equation: $i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$. (ii) Apply Schrödinger's time-independent equation to a particle in a 1D box of length L . Find energy eigenvalues and normalised wave functions. For an electron in a box of width $3 \text{ \AA}$, compute the energies of the first three levels, and find the frequency of the photon emitted in the $n = 3 \rightarrow n = 1$ transition.	3 + 4	4	5
90	(a) Compare classical bits and quantum bits (qubits). For N qubits, how many states can be represented simultaneously? What is superposition?	2	5	2
	(b) Write the matrix representations of Pauli matrices $\sigma_x, \sigma_y, \sigma_z$ (X, Y, Z gates). Verify that each is both unitary and Hermitian.	2	5	3
	(c) For the following cases, find $?\rangle$ showing all steps: (i) $ 0\rangle \xrightarrow{H} \xrightarrow{X} \xrightarrow{H} ?\rangle$ (ii) $ 1\rangle \xrightarrow{H} \xrightarrow{X} \xrightarrow{H} ?\rangle$ (iii) If a single-qubit gate G applied 4 times in series always restores the input state $ 0\rangle \rightarrow 0\rangle$ and $ 1\rangle \rightarrow 1\rangle$, identify all possible gates G .	3	5	4
	(d) Find the output $ \phi_1\phi_2\rangle$ for all possible 2-qubit inputs $ \psi_1\psi_2\rangle$ in the circuit below. Present results in a Truth Table. State whether any output is an entangled state and justify. $ \psi_1\rangle - [H] - \bullet - [H] - \phi_1\rangle$ $ \psi_2\rangle - [X] - \oplus - [X] - \phi_2\rangle$	7	5	5
	OR			
	(e) Find the output for all possible 2-qubit inputs $ \psi_1\psi_2\rangle$ in the circuit below. Present results and explain the output in each case. $ \psi_1\rangle - [H] - \bullet - \phi_1\rangle$ $ \psi_2\rangle - [H] - \oplus - [H] - \phi_2\rangle$ Also prove using matrix multiplication that $H^2 = I$ (Hadamard is its own inverse). Show that $X^2 = Y^2 = Z^2 = I$ as well.	7	5	5

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SAMPLE PAPER – SET 3

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Time: 3 Hours

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Q.No	QUESTION	Marks	CO	BL
90	(a) Explain the terms: (i) Active medium, (ii) Population inversion, (iii) Metastable state, and (iv) Optical resonator. What role does each play in laser action?	2	1	2
	(b) Match the following: (i) Population inversion (α) Converts active medium into light oscillator (ii) Optical resonator (β) Necessary condition for lasing (iii) Active medium (Ruby) (γ) Al_2O_3 with Cr^{3+} ions (iv) 3-level laser (δ) Gives pulsed output (Ruby laser)	2	1	1
	(c) (i) Explain four-level pumping. Why is population inversion always easily maintained in a four-level laser system compared to a three-level system? (ii) State and explain the condition for light amplification in a laser system.	3	1	3
	(d) (i) Derive the expression: $\frac{R_{stim}}{R_{spont}} = \frac{B_{21}\rho(\nu)}{A_{21}} = \frac{1}{e^{h\nu/k_B T} - 1}$. Comment on the significance of this result for visible light at room temperature. (ii) A He-Ne laser tube has length 0.50 m. Calculate the number of longitudinal modes that can oscillate at $\lambda = 632.8$ nm. [Use $\Delta\nu = c/2L$, mode spacing $= c/2L$.] (iii) A pulsed Ruby laser emits 2 J of energy in a 5 ns pulse at $\lambda = 694.3$ nm. Find the peak power and estimate the number of photons in the pulse.	3 + 2 + 2	1	4
OR				
	(e) (i) With the help of a neat schematic and energy-level diagram, explain the construction and working of the He-Ne laser. Describe the energy collision transfer mechanism between He and Ne atoms and identify the transition responsible for the 6328 Å output. (ii) List any four characteristic properties of the He-Ne laser output and state four applications of He-Ne lasers.	4 + 3	1	3
90	(a) Define: (i) Acceptance cone, (ii) Fractional refractive index change $\Delta = (\mu_1 - \mu_2)/\mu_1$, and (iii) V-number. How does the V-number determine the mode of operation?	2	2	2
	(b) An optical fibre has $n_{core} = 1.60$ and $n_{clad} = 1.50$. A ray is launched from air at $\theta_i = 20^\circ$. Does TIR occur inside the fibre? Verify with complete calculation.	2	2	3
	(c) Compare Intrinsic and Extrinsic fibre-optic sensors. Give the working principle, two examples, and two advantages for each type.	3	2	3
	(d) (i) Derive the expression for V-number of an optical fibre: $V = \frac{2\pi a}{\lambda} \sqrt{\mu_1^2 - \mu_2^2}$. How does V determine whether the fibre is single-mode or multimode? What is the cutoff condition? (ii) A step-index fibre has $n_1 = 1.48$, $n_2 = 1.46$, core radius $a = 4 \mu\text{m}$, $\lambda = 1550$ nm. Calculate (a) NA, (b) V-number, (c) state mode of operation. (iii) If the initial power in a fibre is 2 mW and after 5 km it is 0.4 mW, find the attenuation in dB/km.	3 + 2 + 2	2	4

Q.No	QUESTION	Marks	CO	BL
OR				
(e)	(i) Draw the block diagram of a fibre-optic communication system. Explain the function of each block (optical transmitter, communication channel, optical receiver). (ii) Show that when the fibre is immersed in a medium of refractive index $n_0 > 1$, $NA = \sqrt{\mu_1^2 - \mu_2^2}/n_0$, and hence $NA > 1$ is possible when $n_0 > 1$. What does this physically imply? (iii) List and briefly explain any four advantages of optical fibre communication over metallic-wire communication.	2 + 3 + 2	2	3
90	(a) State the two postulates of STR. Using the second postulate alone, demonstrate that simultaneity is a relative concept (i.e., events simultaneous in one frame may not be simultaneous in another).	2	3	3
	(b) Find the kinetic energy of an electron moving at $v = 0.85c$ using: (i) classical formula and (ii) relativistic formula. Calculate the percentage error in the classical result.	2	3	4
	(c) A spaceship of proper length 50 m moves at $0.6c$ relative to Earth and fires a rocket at $0.8c$ relative to itself (forward). (i) What is the rocket's speed relative to Earth? (ii) What length does an Earth observer measure for the spaceship?	3	3	4
	(d) (i) Using Lorentz transformation, derive the relativistic velocity transformation equations $u'_x = (u_x - v)/(1 - u_x v/c^2)$ and $u'_y = u_y/[\gamma(1 - u_x v/c^2)]$. Discuss the limit $u_x = c$. (ii) A particle of rest mass m_0 has kinetic energy equal to its rest-mass energy ($K = m_0 c^2$). Find its velocity. (iii) Two events occur at the same location in frame S , separated by time interval 5 s. Frame S' moves at $0.6c$ relative to S . What is the spatial separation of the events in S' ?	3 + 2 + 2	3	4
OR				
	(e) (i) Assuming that the transformation between two inertial frames must be linear, use the two postulates of STR to derive the complete Lorentz transformation equations. (ii) Derive the mass-energy equivalence $E = mc^2$ from the work-energy theorem using relativistic force and momentum. Also derive $E^2 = p^2 c^2 + m_0^2 c^4$ and apply it to a photon.	3 + 4	3	4
90	(a) In Compton scattering, why is the effect not observable with visible light but significant for X-rays? Estimate the fractional Compton shift $\Delta\lambda/\lambda$ for visible light ($\lambda = 5000 \text{ \AA}$) at $\theta = 90^\circ$ and compare with that for X-rays ($\lambda = 0.5 \text{ \AA}$).	2	4	3
	(b) A particle is in thermal equilibrium at temperature $T = 300 \text{ K}$ with average $KE = \frac{3}{2} k_B T$. Find the de Broglie wavelength for: (i) an electron and (ii) a neutron ($m_n = 1.67 \times 10^{-27} \text{ kg}$).	2	4	3
	(c) (i) Using HUP, estimate the minimum energy of a neutron confined in a nucleus of size 10^{-15} m . Comment on whether electrons can exist inside the nucleus. (ii) Compute the lowest three energy levels of an electron in a box of width $3 \text{ \AA}$. Draw the energy level diagram.	3	4	5

Q.No	QUESTION	Marks	CO	BL
(d)	(i) Derive Schrödinger's time-independent wave equation from the classical wave equation using the de Broglie relation and the energy conservation equation $E = K + V$. (ii) For a particle in a 1D box (width a , n -th state), find $\langle x \rangle$, $\langle x^2 \rangle$, $\langle p \rangle$, and $\langle p^2 \rangle$. Use these to verify Heisenberg's Uncertainty Principle: $\Delta x \Delta p \geq \hbar/2$. (iii) A particle of mass 9.1×10^{-31} kg is confined in a box of width 10^{-10} m. Calculate the zero-point energy ($n = 1$).	3 + 2 + 2	4	5
OR				
(e)	(i) Describe the Compton scattering experiment with a neat diagram. Derive the Compton wavelength shift formula $\Delta\lambda = \frac{h}{m_0c}(1 - \cos\theta)$ using conservation of energy and momentum. (ii) State all properties of a well-behaved wave function. If $\psi(x) = A \sin(n\pi x/a)$ for $0 < x < a$, normalise it and find the probability of locating the particle in $0 < x < a/4$ for $n = 1$.	4 + 3	4	5
(a)	Construct a NAND gate using AND and NOT. Then show that NAND is a universal gate by implementing AND, OR, and NOT using only NAND gates.	2	5	3
(b)	Explain quantum superposition. For $ \psi\rangle = \alpha 0\rangle + \beta 1\rangle$, state the normalization condition. What happens to the superposition upon measurement?	2	5	2
UNIT 5: QUANTUM COMPUTING				
(c)	A gate G is connected in series. When applied an even number of times, the input is restored; when applied an odd number of times: $ 0\rangle \rightarrow 0\rangle$, $ 1\rangle \rightarrow (-1)^n 1\rangle$ (for n applications). Identify all single-qubit gates G satisfying this condition. Justify using matrix multiplication.	3	5	5
(d)	Solve the following 2-qubit quantum circuit for all four possible inputs $ 00\rangle, 01\rangle, 10\rangle, 11\rangle$. Show all matrix steps. Present the outputs in a Truth Table and state whether each output is a Bell state: $ \psi_1\rangle - [H] - \bullet - \phi_1\rangle$ $ \psi_2\rangle - [X] - \oplus - [Z] - \phi_2\rangle$	7	5	5
OR				
(e)	Consider the circuit: $ \psi_1\rangle - [H] - \bullet - [H] - \phi_1\rangle$ $ \psi_2\rangle \quad \oplus - [X] - \phi_2\rangle$ (i) Evaluate the output for all four inputs $ 00\rangle, 01\rangle, 10\rangle, 11\rangle$. (ii) Using matrix method, show that $\text{SWAP} = \frac{I \otimes I + X \otimes X + Y \otimes Y + Z \otimes Z}{2}$ by explicit matrix computation of the right-hand side.	4 + 3	5	5

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SAMPLE PAPER – SET 4

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Q.No	QUESTION	Marks	CO	BL
90UNIT 1: LASERS	(a) Draw and explain the energy-level diagram showing absorption, spontaneous emission, and stimulated emission. State the rate equations R_{abs} , R_{spon} , R_{stim} in terms of Einstein's coefficients.	2	1	2
	(b) Match the following: (i) S.N. Bose (α) Invented the Ruby laser (ii) T.H. Maiman (β) Bose-Einstein statistics for photons (iii) G.S. Agarwal (γ) Quantum optics and coherence theory (iv) CKN Patel (δ) Invented the CO ₂ laser	2	1	1
	(c) (i) Explain optical pumping and electric discharge pumping with one example each. Which is used in Ruby laser and which in He-Ne laser? (ii) What is the significance of the metastable state in maintaining population inversion? How long does an atom typically stay in a metastable state?	3	1	3
	(d) (i) For a laser system at thermal equilibrium, show that the ratio of stimulated-to-spontaneous emission rate is $R_{stim}/R_{spon} = 1/(e^{h\nu/k_B T} - 1)$. Evaluate this at $T = 6000$ K for $\lambda = 500$ nm. (ii) A Ruby laser emits at $\lambda = 694.3$ nm with average power 10 kW in pulses of duration 10^{-8} s. Find: (a) energy per pulse and (b) number of photons per pulse. (iii) Calculate the angular spread of a He-Ne laser beam ($\lambda = 6328$ Å, beam diameter $D = 2$ mm) using $2\theta = 0.637\lambda/D$.	3 + 2 + 2	1	4
	OR			
	(e) Explain in full detail the construction, working principle, and energy-level diagram of the He-Ne laser . Cover: (i) active medium and mixing ratio, (ii) pumping mechanism (electrical discharge), (iii) energy collision transfer from He to Ne, (iv) the 3s→2p transition responsible for 6328 Å output, (v) why Ne atoms in the 2p level relax quickly to the ground state, and (vi) why He-Ne gives continuous output while Ruby gives pulsed output.	7	1	3
90UNIT 2: FIBRE OPTICS	(a) Why must the refractive index of the cladding be lower than that of the core in an optical fibre? What would happen physically if $\mu_{clad} > \mu_{core}$?	2	2	2
	(b) A step-index fibre has $\mu_1 = 1.62$ and $\mu_2 = 1.52$. Calculate: (i) critical angle, (ii) numerical aperture, and (iii) acceptance angle.	2	2	3
	(c) Write a detailed note on the contributions of any two of the following to photonics/optical communication: (i) N.S. Kapany (ii) S.N. Bose (iii) G.S. Agarwal	3	2	2
	(d) (i) With a neat ray diagram, derive the expression for NA and acceptance angle θ_{max} at the input end of an optical fibre. Apply Snell's law at both the air-core interface and the core-cladding interface. (ii) A multimode fibre has $n_1 = 1.50$, $n_2 = 1.48$, core radius $a = 25$ μm, $\lambda = 850$ nm. Find: (a) V-number, (b) number of modes, (c) NA. (iii) If power launched into a fibre is 1 mW and drops to 0.1 mW after 10 km, calculate the attenuation in dB/km.	3 + 2 + 2	2	4

Q.No	QUESTION		Marks	CO	BL
OR					
(e)	(i) Discuss the different types of losses in optical fibre with emphasis on Rayleigh scattering and absorption. Why is 1550 nm the preferred wavelength for long-haul optical communication? (ii) Draw the refractive-index profiles and explain mode structures for: (a) step-index single mode, (b) step-index multimode, and (c) graded-index multimode fibres. Which has least intermodal dispersion and why?	3 + 4	2	3	
90	(a)	A clock at rest ticks every 1 s. (i) At what speed must it travel so that a stationary observer sees it tick every 2 s? (ii) At what speed does the clock appear to tick every 10 s?	2	3	3
	(b)	Explain the concept of relativistic mass variation with velocity. Derive and sketch m vs v/c . Why does it seem impossible to accelerate any material body to the speed of light?	2	3	3
	(c)	Two observers: A (stationary) and B (moving at $0.9c$) both measure the same moving train. If A measures the train's length as 400 m, what does B measure? If B measures a time interval of 10 s on the train, what does A measure?	3	3	4
	(d)	(i) From the two postulates of STR, derive the complete Lorentz transformation equations for two frames S and S' moving along the x -axis at relative velocity v . (ii) Show that the space-time interval $\Delta s^2 = c^2\Delta t^2 - \Delta x^2$ is Lorentz-invariant. (iii) A particle of rest mass m_0 has kinetic energy $K = 3m_0c^2$. Find its: (a) total energy, (b) velocity v , and (c) momentum p .	3 + 2 + 2	3	4
OR					
(e)	(i) Write down the Galilean transformation equations. At what velocities do they fail? Why? (ii) Derive the relativistic velocity transformation equations from Lorentz transformations and show that the speed of light c is the absolute upper limit for any material body. (iii) An electron and a positron, each moving in opposite directions with kinetic energy equal to $m_e c^2$, undergo pair annihilation. Find the wavelengths of the two gamma photons produced. [$m_e c^2 = 0.511$ MeV]	2 + 3 + 2	3	4	
90	(a)	A light source emits photons at $\lambda = 630$ nm with power 6 W. An atom in an excited state has a lifetime of 50 ps. Find: (i) number of photons emitted per second, and (ii) minimum uncertainty in the energy of the atom.	2	4	3
	(b)	An electron and a photon both have kinetic energy 1 eV. Compare their de Broglie wavelengths. Which is larger and why?	2	4	4
	(c)	Derive the Compton wavelength shift formula $\Delta\lambda = \frac{h}{m_0c}(1 - \cos\theta)$. Find $\Delta\lambda$ when $\theta = 0^\circ, 90^\circ$, and 180° . What is the Compton wavelength of an electron?	3	4	4
	(d)	(i) Using Schrödinger's time-independent equation for a particle in a 1D infinite potential well of width a , apply boundary conditions to show that energy is quantised: $E_n = n^2h^2/8ma^2$. Derive the normalised wave functions. (ii) For an electron in a well of width $a = 2 \times 10^{-10}$ m, calculate the energies of the first three levels. Draw the energy-level diagram. (iii) Find the probability of locating the particle in the ground state ($n = 1$) in the region $0 < x < a/3$.	3 + 2 + 2	4	5
UNIT 4: QUANTUM THEORY					

Q.No	QUESTION	Marks	CO	BL
OR				
(e)	(i) What is the physical significance of the normalisation condition? If $\psi = Ae^{-i(Et/\hbar + \alpha x/2)}$ is normalised over $[0, L]$, find A . (ii) Using Schrödinger's time-independent wave equation, obtain normalised wave functions for a particle in a 1D infinite potential well of width a . Plot $ \psi_n ^2$ for $n = 1, 2, 3$, identify nodes and antinodes, and locate positions of maximum probability density for each level.	2 + 5	4	5
(a)	List at least four key limitations of classical digital computers. Explain any two in detail. How do quantum computers overcome these limitations using superposition and entanglement?	2	5	2
(b)	Write truth tables for AND, OR, NAND, and XOR gates. Using a truth table, show that a single NAND gate with both inputs connected implements a NOT gate.	2	5	2
UNIT 5: QUANTUM COMPUTING				
(c)	Find $?\rangle$ in the following cases (show all matrix steps): (i) $ 0\rangle \xrightarrow{H} \bullet \xrightarrow{H} ?\rangle$; $ 0\rangle \longrightarrow \oplus \longrightarrow ?\rangle$ (CNOT applied) (ii) $ 0\rangle \xrightarrow{H} \xrightarrow{Z} \xrightarrow{H} ?\rangle$ (Hint: relate (i) and (ii) – are these circuits equivalent?)	3	5	4
(d)	The circuit below is the standard Bell-state generator. Solve for all four possible 2-qubit inputs $ \psi_1\psi_2\rangle \in \{ 00\rangle, 01\rangle, 10\rangle, 11\rangle\}$. Present outputs $ \phi_1\phi_2\rangle$ in a Truth Table. Identify all four Bell states by name and write their mathematical expressions. $ \psi_1\rangle - [H] - \bullet - \phi_1\rangle$ $ \psi_2\rangle \quad \oplus \quad \phi_2\rangle$	7	5	5
OR				
(e)	Solve the following 2-qubit circuit for all possible inputs showing all matrix steps. Present a Truth Table and explain whether the outputs are entangled: $ \psi_1\rangle - [H] - \bullet - \phi_1\rangle$ $ \psi_2\rangle - [H] - \oplus - [X] - \phi_2\rangle$ Also, using the 4×4 CNOT matrix, verify that CNOT flips the target qubit only when the control qubit is $ 1\rangle$, for each input case.	7	5	5

CO – Course Outcome. BL – Bloom's Taxonomy: 1-Remember, 2-Understand, 3-Apply, 4-Analyse, 5-Evaluate, 6-Create.